MAT 3007 – Optimization

The Simplex Tableau

Lecture 10

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Recap

Simplex Method: Neighborng BFS

Neighbors and Principle Idea:

- Two basic solutions are neighbors if they differ in exactly one basic index.
- Basic idea: Move from one BFS to a neighboring BFS while improving the objective function.

How to Move to a Neighbor:

- Move along j-th basic directions from a BFS $\boldsymbol{x}$: e_k with $j = N(k)$

$$\boldsymbol{x} \to \boldsymbol{x} + \theta \boldsymbol{d}, \quad \boldsymbol{d} = [\boldsymbol{d}_B; \boldsymbol{d}_N] = [-\boldsymbol{A}_B^{-1}\boldsymbol{A}_j; 0; \ldots; 1; \ldots; 0].$$

- The definition of basic direction guarantees feasibility $\boldsymbol{A}(\boldsymbol{x} + \theta \boldsymbol{d}) = \boldsymbol{b}$ for all θ. Therefore, we need to choose θ to ensure $\boldsymbol{x} + \theta \boldsymbol{d} \geq \boldsymbol{0}$.

Reduced Costs & Selecting Directions:

- The reduced cost for any j is defined as

Note $\bar{c}_B = 0$

$$\bar{c}_j = c_j - \boldsymbol{c}_B^\top \boldsymbol{A}_B^{-1} \boldsymbol{A}_j = \boldsymbol{c}^\top \boldsymbol{d}.$$

- Moving from $\boldsymbol{x}$ to $\boldsymbol{y} = \boldsymbol{x} + \theta \boldsymbol{d}$, where $\boldsymbol{d}$ is j-th basic direction, reduces the objective by $\theta \bar{c}_j$.

Theorem: Stopping Criterion

Let $\boldsymbol{x}$ be a BFS associated with the basis $B(1), \ldots, B(m)$ and let $\bar{\boldsymbol{c}}$ be the corresponding vector of reduced costs. If $\bar{\boldsymbol{c}} \geq 0$, then $\boldsymbol{x}$ must be optimal.

- It means that if we can not find a neighboring BFS that improves the objective function, then the current BFS is an optimal solution.

Choose a Stepsize: Minimum-Ratio-Test

Specifying θ:

- Select j-th basic direction $\boldsymbol{d}$ with $\bar{c}_j < 0$. Move along $\boldsymbol{d}$ as much as possible:

$$\theta^* = \max\{\theta \geq 0 : \boldsymbol{x} + \theta \boldsymbol{d} \geq \boldsymbol{0}\}.$$

We consider two cases:

1. If $\boldsymbol{d} \geq \boldsymbol{0}$, then $\theta^* = \infty$. The original LP is unbounded.
2. If $d_i < 0$ for some i, then we can solve:

$$\theta^* = \min_{\{i : d_i < 0\}} -\frac{x_i}{d_i} = \min_{\{i \in B : d_i < 0\}} -\frac{x_i}{d_i}.$$

Moving to a New Basis

Assume θ^* is finite (otherwise the problem is unbounded), then we can move to another feasible point:

$$\boldsymbol{y} = \boldsymbol{x} + \theta^* \boldsymbol{d}.$$

Let $B(\ell) \in B$ be an index with $\theta^* = -x_{B(\ell)}/d_{B(\ell)}$.

$$y_{B(\ell)} = x_{B(\ell)} + \theta^* d_{B(\ell)} = 0.$$

Thus, the basic variable $x_{B(\ell)}$ has become zero, whereas the non-basic variable x_j is now positive (equal to θ^*). The basis has changed to: (It remains to prove the correpsonding columns of A linearly independent.)

$$B(1), \ldots, B(\ell-1), j, B(\ell+1), \ldots, B(m).$$

Properties of the New BFS

Theorem: Properties of $\boldsymbol{y}$

Let $\boldsymbol{x}$ be a nondegenerate BFS ($\boldsymbol{x}_B > 0$) with basic indices B and let $\boldsymbol{y} = \boldsymbol{x} + \theta^* \boldsymbol{d}$ be generated by the simplex iteration. Then, $\boldsymbol{y}$ is a basic feasible solution associated with the basic indices

$$\{B(1), \ldots, B(\ell - 1), j, B(\ell + 1), \ldots, B(m)\}.$$

An Iteration of the Simplex Method

Initialization: We start from a BFS $\boldsymbol{x}$ (with basic set B).

1. Compute the reduced costs $\bar{\boldsymbol{c}}$:
$$\bar{c}_j = c_j - \boldsymbol{c}_B^\top \boldsymbol{A}_B^{-1} \boldsymbol{A}_j.$$
   - If none of the reduced costs is negative, then $\boldsymbol{x}$ is optimal.
   - Otherwise choose some j such that $\bar{c}_j < 0$.
2. Compute the jth basic direction
$$\boldsymbol{d} = [-\boldsymbol{A}_B^{-1}\boldsymbol{A}_j; 0; \ldots; 1; \ldots; 0]$$
   - If $\boldsymbol{d} \geq \boldsymbol{0}$, then the problem is unbounded
   - Otherwise, compute $\theta^* = \min_{i \in B, d_i < 0} \{-\frac{x_i}{d_i}\}$.
3. Set $\boldsymbol{y} = \boldsymbol{x} + \theta^* \boldsymbol{d}$. Then, $\boldsymbol{y}$ is the new BFS with index j replacing $B(\ell)$ in the basis, where $B(\ell)$ is the index attaining the minimum in θ^*. The objective function value is changed by $\theta^* \boldsymbol{c}^\top \boldsymbol{d} = \theta^* \bar{c}_j$.
4. Repeat these procedures.

Degeneracy

We call a basic feasible solution $\boldsymbol{x}$ (non-)degenerate if (none) some of the basic variables are 0.

$$\theta^* = \min_{\{i \in B \mid d_i < 0\}} -\frac{x_i}{d_i}.$$

If for some $i \in B$ with $x_i = 0$ and $d_i < 0$, then we obtain $\theta^* = 0$. Hence $\boldsymbol{y} = \boldsymbol{x}$ and the objective value reamins unchanged.

Change of basis: We still replace the basic index $B(\ell)$ by j and proceed to the next iteration.

- There might be a cycle, i.e., visit the same BFS more than once. We need a proper pivoting rule to avoid it.

Theorem: Bland's Rule

If we use the smallest index rule for choosing both the entering basis and the exiting basis, then no cycle will occur in the simplex algorithm.

Bland's Rule

- Choose the smallest index j in $\{j : \bar{c}_j < 0\}$ to enter the basis.
- Choose the smallest index $B(\ell)$ that satisfies $d_{B(\ell)} < 0$ and $\theta^* = -x_{B(\ell)}/d_{B(\ell)}$ to exit the basis.

with this rule, if there is a tie between B(ℓ) and B(k), we will choose the smaller between B(ℓ) and B(k) rather than the smaller between ℓ and k.

Finding an Initial BFS

How to find an initial BFS?

with $b \geq 0$

$Ax \leq b \iff Ax + s = b$ then $\begin{pmatrix} x \\ s \end{pmatrix} = \begin{pmatrix} 0 \\ b \end{pmatrix}$

$x \geq 0$ $\quad x \geq 0, s \geq 0$ is a BFS

- We have a trivial BFS if the standardd form is derived by adding slack variables to each constraint.
- Two-Phase simplex method
- Big-M method

Two-Phase Simplex Method

In the two-phase simplex method, we first solve an auxiliary problem ($\mathbb{1}$ is an all-one vector):

$$\begin{aligned} \text{minimize}_{\boldsymbol{x},\boldsymbol{y}} \quad & \mathbb{1}^\top \boldsymbol{y} \\ \text{subject to} \quad & \boldsymbol{A}\boldsymbol{x} + \boldsymbol{y} = \boldsymbol{b} \\ & \boldsymbol{x}, \boldsymbol{y} \geq \boldsymbol{0}. \end{aligned}$$

Without loss of generality, we can assume $\boldsymbol{b} \geq \boldsymbol{0}$ (otherwise, we pre-multiply that row by -1).

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- Trivial BFS for this auxiliary problem: $\boldsymbol{x} = \boldsymbol{0}$, $\boldsymbol{y} = \boldsymbol{b} \geq \boldsymbol{0}$.
- We can apply the simplex method to solve it.

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Theorem: Feasibility

The original problem is feasible if and only if the optimal value of the auxiliary problem is 0.

Two-Phase Simplex Method

Hence, we can solve the auxiliary problem by the simplex method:

1. If the optimal value is not 0, then we can claim that the original problem is infeasible.
2. If the optimal value is 0, then we can find an optimal BFS $(\boldsymbol{x}^*, \boldsymbol{y}^* = \mathbf{0})$ with basis B to the auxiliary problem.

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   - If $\boldsymbol{x}^*$ contains the full basis B, then x^* is a BFS for the orignal problem with basis B;

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   - If $\boldsymbol{x}^*$ contains the full basis B, then x^* is a BFS for the orignal problem with basis B;
   - If $\boldsymbol{x}^*$ does not contains the full basis B, denote those basic indexes in $\boldsymbol{x}^*$ as $B(1), ..., B(k)$ with $k < m$. Then we can find a set $S = \{S(1), ..., S(m-k)\}$ disjoint from B such that columns of $A_{B \cup S}$ are linearly independent. Therefore, $B \cup S$ is a basis for $\boldsymbol{x}^*$ in the original problem.

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$\rightsquigarrow$ We can start from $\boldsymbol{x}^*$ and its basis to initialize the original simplex method.

Procedure of the Two-Phase Method

Phase I:

1. Construct the auxiliary problem such that $\boldsymbol{b} \geq \mathbf{0}$.
2. Solve the auxiliary problem using the simplex method.
   - If we reach an optimal solution with optimal value greater than 0, then the original problem is infeasible.
3. If the optimal value is 0 with an optimal solution $(\boldsymbol{x}^*, \boldsymbol{y}^*)$, then we enter phase II.

Phase II:

Solve the original problem starting from the BFS $\boldsymbol{x}^*$:

- If $(\boldsymbol{x}^*, \boldsymbol{y}^*)$ is degenerate for the auxiliary problem, then we need to supplement some indices to have a basis for the original problem.

The Big-M Method

There is another method that can be used to solve LP without a starting BFS. Consider the following auxiliary problem:

$$\begin{aligned} \text{minimize} \quad & \boldsymbol{c}^\top \boldsymbol{x} + M \sum_{i=1}^{m} y_i \\ \text{subject to} \quad & \boldsymbol{A}\boldsymbol{x} + \boldsymbol{y} = \boldsymbol{b} \\ & \boldsymbol{x}, \boldsymbol{y} \geq \boldsymbol{0}. \end{aligned}$$

This problem has an initial BFS $\boldsymbol{x} = \boldsymbol{0}, \boldsymbol{y} = \boldsymbol{b} \geq \boldsymbol{0}$ (assuming $\boldsymbol{b} \geq \boldsymbol{0}$). $\rightsquigarrow$ We can use the simplex method to solve it.

In the simplex procedure, pretend that M is a very large value (larger than any specified number):

- If the original problem is feasible, then the optimal solution will have $\boldsymbol{y}^* = 0$ and does not involve basic indices in $\boldsymbol{y}$.

The Simplex Tableau

Simplex Tableau

- Let B the current basis with basis matrix $\boldsymbol{A}_B$ and basis objective coefficients $\boldsymbol{c}_B$.
- The simplex tableau is a table with the following structure:

$\boldsymbol{c}^\top - \boldsymbol{c}_B^\top \boldsymbol{A}_B^{-1} \boldsymbol{A}$	$-\boldsymbol{c}_B^\top \boldsymbol{A}_B^{-1} \boldsymbol{b}$
$\boldsymbol{A}_B^{-1} \boldsymbol{A}$	$\boldsymbol{A}_B^{-1} \boldsymbol{b}$

Simplex Tableau – the Lower Section

$\boldsymbol{c}^\top - \boldsymbol{c}_B^\top \boldsymbol{A}_B^{-1} \boldsymbol{A}$	$-\boldsymbol{c}_B^\top \boldsymbol{A}_B^{-1} \boldsymbol{b}$
$\boldsymbol{A}_B^{-1} \boldsymbol{A}$	$\boldsymbol{A}_B^{-1} \boldsymbol{b}$

The lower section can be viewed as an equivalent transformation of the constraint $\boldsymbol{A}\boldsymbol{x} = \boldsymbol{b}$ to $\boldsymbol{A}_B^{-1}\boldsymbol{A}\boldsymbol{x} = \boldsymbol{A}_B^{-1}\boldsymbol{b}$:

Simplex Tableau – the Lower Section

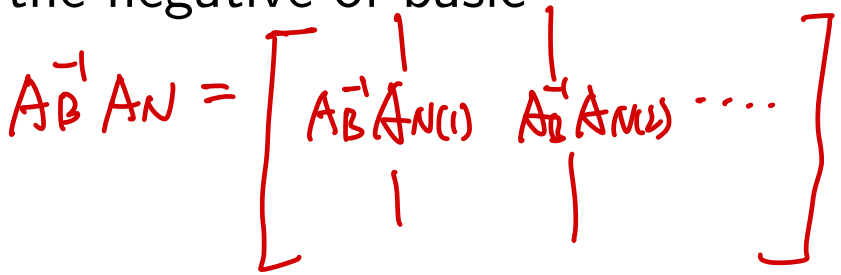
$\boldsymbol{c}^\top - \boldsymbol{c}_B^\top \boldsymbol{A}_B^{-1}\boldsymbol{A}$	$-\boldsymbol{c}_B^\top \boldsymbol{A}_B^{-1}\boldsymbol{b}$
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Lower-Left section: writing $\boldsymbol{A} = [\boldsymbol{A}_B, \boldsymbol{A}_N]$, we have

$$\boldsymbol{A}_B^{-1}\boldsymbol{A} = [\boldsymbol{I}, \boldsymbol{A}_B^{-1}\boldsymbol{A}_N].$$

- This block must contain an identity matrix.
- The columns of $\boldsymbol{A}_B^{-1}\boldsymbol{A}_N$ contain the negative of basic directions, $-\boldsymbol{A}_B^{-1}\boldsymbol{A}_j$.



Simplex Tableau – the Lower Section

$\boldsymbol{c}^\top - \boldsymbol{c}_B^\top \boldsymbol{A}_B^{-1}\boldsymbol{A}$	$-\boldsymbol{c}_B^\top \boldsymbol{A}_B^{-1}\boldsymbol{b}$
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- This block must contain an identity matrix.
- The columns of $\boldsymbol{A}_B^{-1}\boldsymbol{A}_N$ contain the negative of basic directions, $-\boldsymbol{A}_B^{-1}\boldsymbol{A}_j$.

Lower-Right section: contains the current basic variables $\boldsymbol{x}_B = \boldsymbol{A}_B^{-1}\boldsymbol{b}$ and the current BFS is

$$\boldsymbol{x} = [\boldsymbol{x}_B; \boldsymbol{x}_N] = [\boldsymbol{A}_B^{-1}\boldsymbol{b}; \boldsymbol{0}].$$

Simplex Tableau – the Upper Section

row vector $(\bar{c}_1\ \bar{c}_2\ \cdots\ \bar{c}_n)$

$\boldsymbol{c}^\top - \boldsymbol{c}_B^\top \boldsymbol{A}_B^{-1}\boldsymbol{A}$	$-\boldsymbol{c}_B^\top \boldsymbol{A}_B^{-1}\boldsymbol{b}$
$\boldsymbol{A}_B^{-1}\boldsymbol{A}$	$\boldsymbol{A}_B^{-1}\boldsymbol{b}$

Upper-Left section: represent the reduced costs

$$\bar{\boldsymbol{c}}^\top = \boldsymbol{c}^\top - \boldsymbol{c}_B^\top \boldsymbol{A}_B^{-1}\boldsymbol{A}$$

- Recall that $\boldsymbol{c}_B = \boldsymbol{0}$.

Simplex Tableau – the Upper Section

$\boldsymbol{c}^\top - \boldsymbol{c}_B^\top \boldsymbol{A}_B^{-1} \boldsymbol{A}$	$-\boldsymbol{c}_B^\top \boldsymbol{A}_B^{-1} \boldsymbol{b}$
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Upper-Left section: represent the reduced costs

$$\bar{\boldsymbol{c}}^\top = \boldsymbol{c}^\top - \boldsymbol{c}_B^\top \boldsymbol{A}_B^{-1} \boldsymbol{A}$$

- Recall that $\boldsymbol{c}_N = \boldsymbol{0}$.

Upper-Left section: represent the the negative of the current objective value

$$\boldsymbol{c}^\top \boldsymbol{x} = \boldsymbol{c}_B^\top \boldsymbol{x}_B = \boldsymbol{c}_B^\top \boldsymbol{A}_B^{-1} \boldsymbol{b}.$$

Simplex Tableau

Thus, the simplex tableau should look like (after reordering the columns in $[B; N]$):

corresponds to B.

$\mathbf{0}_m^\top$	$\boldsymbol{c}_N^\top - \boldsymbol{c}_B^\top \boldsymbol{A}_B^{-1} \boldsymbol{A}_N$	$-\boldsymbol{c}_B^\top \boldsymbol{x}_B$
$\boldsymbol{I}_m$	$\boldsymbol{A}_B^{-1} \boldsymbol{A}_N$	$\boldsymbol{x}_B$

Here, $\mathbf{0}_m \in \mathbb{R}^m$ is a vector of m zeros and $\boldsymbol{I}_m$ is the m-dimensional identity matrix.

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Here, $\mathbf{0}_m \in \mathbb{R}^m$ is a vector of m zeros and $\boldsymbol{I}_m$ is the m-dimensional identity matrix.

This form of an LP is called the canonical form:

- The constraint matrix for the basic variables (not necessarily the first m columns) is an identity matrix.
- The objective function part for the basic variables is zero.

Example

The production problem

$$
\begin{array}{rrrrrrrl}
\text{minimize} & -x_1 & -2x_2 & & & & & \\
\text{subject to} & x_1 & & +s_1 & & & & = 100 \\
& & 2x_2 & & +s_2 & & & = 200 \\
& x_1 & +x_2 & & & +s_3 & & = 150 \\
& x_1, & x_2, & s_1, & s_2, & s_3 & & \geq 0
\end{array}
$$

It is already in the canonical form as follows:

$\bar{c}_j = c_j - c_B^T A_B^{-1} A_j = c_j \rightarrow$ upper-left

function value $= 0 \rightarrow$ upper-right

$A_B^{-1} A = A \rightarrow$ lower-left

$A_B^{-1} b = b \rightarrow$ lower-right

-1	-2	0	0	0	0
1	0	1	0	0	100
0	2	0	1	0	200
1	1	0	0	1	150

$\begin{pmatrix} x_1 \\ x_2 \\ s_1 \\ s_2 \\ s_3 \end{pmatrix} = \begin{pmatrix} 0 \\ 0 \\ 100 \\ 200 \\ 150 \end{pmatrix}$ is an initial BFS.

$B = \{3, 4, 5\}$

$A_B = I_3$

- Are we already in an optimal solution?

  $\rightsquigarrow$ No.

Simplex Tableau

The simplex tableau will maintain a canonical form for each iteration until we reach optimality:

- For now we assume that we already have a canonical form to start with (an initial BFS).

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Task: We want to map each algebraic step of the simplex method to an operation on the simplex tableau:

1. Compute the reduced costs.
2. Choose the incoming basic index.
3. Compute θ^* and choose the outgoing basic index.
4. Update the tableau with the new basis.

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Task: We want to map each algebraic step of the simplex method to an operation on the simplex tableau:

1. Compute the reduced costs.
2. Choose the incoming basic index.
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We call the procedure of transforming from one canonical form (one set of basis) to another one pivoting.

Pivoting Step I: Choose the Incoming Basis

The reduced costs are the coefficients in the upper-left corner of the tableau.

- Choose an column with negative reduced cost to be the incoming basic index.

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Consider the example (production plan):

	x_1	x_2	x_3	x_4	x_5	
	-1	-2	0	0	0	0
B(1)	1	0	1	0	0	100
B(2)	0	2	0	1	0	200
B(3)	1	1	0	0	1	150

Look at upper-left, i.e. reduce cost.

current x_B

negative of 1-st basic direction

We can choose either the first or second column as the entering basic index (if we use Bland's rule, then we choose the first one).

Pivoting Step II: θ^* and Outgoing Basis

Assume we have chosen column j as the incoming basis and $\boldsymbol{d}$ is j-th basic direction. The step size θ^* was determined via:

$$\theta^* = \min_{d_i<0, i\in B} -x_i/d_i = \min_i \left\{ \frac{\bar{b}_i}{\bar{A}_{ij}} : \bar{A}_{ij} > 0 \right\}$$

where $\bar{\boldsymbol{b}}$ is the lower-right section of the tableau and $\bar{\boldsymbol{A}}$ is the lower-left section of the tableau.

- This is called the Minimal Ratio Test (MRT).

Pivoting Step II: θ^* and Outgoing Basis

If $\bar{A}_{ij} \leq 0$ for all i, then the problem is unbounded. ($\rightsquigarrow d \geq 0$).

Otherwise, assume an index i achieves the minimum in:

$$\theta^* = \min_i \left\{ \frac{\bar{b}_i}{\bar{A}_{ij}} : \bar{A}_{ij} > 0 \right\}$$

- We call row i the pivot row.
- The column $B(i)$ leaves the basis. A trick to recognize $B(i)$: the column in the current basis whose i-th element is 1.

Example: If we choose column 1 to be the incoming basis:

			x_3	x_4	x_5	
B	-1	-2	0	0	0	0
3	1	0	1	0	0	100
4	0	2	0	1	0	200
5	1	1	0	0	1	150

$100/1 = 100$

$150/1 = 150$

MRT will choose the third column to be the outgoing basis.

Example – Continued

B	-1	-2	0	0	0	0
3	1	0	1	0	0	100
4	0	2	0	1	0	200
5	1	1	0	0	1	150

200/2 = 100

150/1 = 150

What if we choose column 2 to be the incoming basis?

- The MRT will choose the fourth column as outgoing basis.
- The MRT finds that the second row achieves the minimum ratio. Then we choose the basis whose second row element is 1 to be the outgoing basis, which is column 4.

Terminologies

- We call the incoming column the pivot column.
- We call the row that achieves the MRT the pivot row (determines the outgoing basis).
- The intersection element of the pivot column and the pivot row is called the pivot element.

B	-1	-2	0	0	0	0
3	1	0	1	0	0	100
4	0	2	0	1	0	200
5	1	1	0	0	1	150

$j=1$ replaces $B(1)=3$ in the basis. In the new basis $\overline{B}=\{1,4,5\}$.

$\vec{A}_{\overline{B}} A_j = \begin{pmatrix} 1 \\ 0 \\ 0 \end{pmatrix}$

we want it to be $\begin{pmatrix} 1 \\ 0 \\ 0 \end{pmatrix}$ for lower left and 0 for upper-left

Updating the Tableau

Assume we have determined the incoming and outgoing basis (pivot element $\bar{A}_{ij}$).

We perform the following two steps:

1. Divide each element in the pivot row by the pivot element.
2. Add proper multiples (could be negative) of the pivot row (after the first step) to each other rows, including the top row of objective coefficients, such that all other elements in the pivot column become zeros (including the top row).

- Both operations include the right-hand-side column of $\bar{b}$.

Afterwards, the new pivot column should be $(0; \ldots; 0; 1; 0; \ldots; 0)$ with 1 at the pivot row.

- The new resulting tableau will still be in a canonical form, however, with the new basis.

Simplex Method in the Tableau

We have shown how to get from one canonical form to another, we then repeat this procedure until we reach optimality.

- When choosing the incoming and the outgoing basis, we use the smallest index rule.
- This will guarantee that the simplex iterations will terminate in a finite number of steps.

We also attach the index of the basis to the left of the tableau to indicate the current basis (just for clarity).

Example: Iteration I

Consider the production example. The initial simplex tableau is ($B = \{3, 4, 5\}$):

B	-1	-2	0	0	0	0
3	1	0	1	0	0	100
4	0	2	0	1	0	200
5	1	1	0	0	1	150
	0	1	-1	0	1	50

We use the smallest index rule. The pivot column (incoming basis) is the first column, the pivot row is the first row (outgoing basis is column 3), the pivot element is 1 (in yellow).

- Divide the pivot row by the pivot element.
- Add proper multiples of row 1 to other rows (including the top row) such that all other elements in the new pivot column (except the pivot element) become zero.

Example: Iteration II

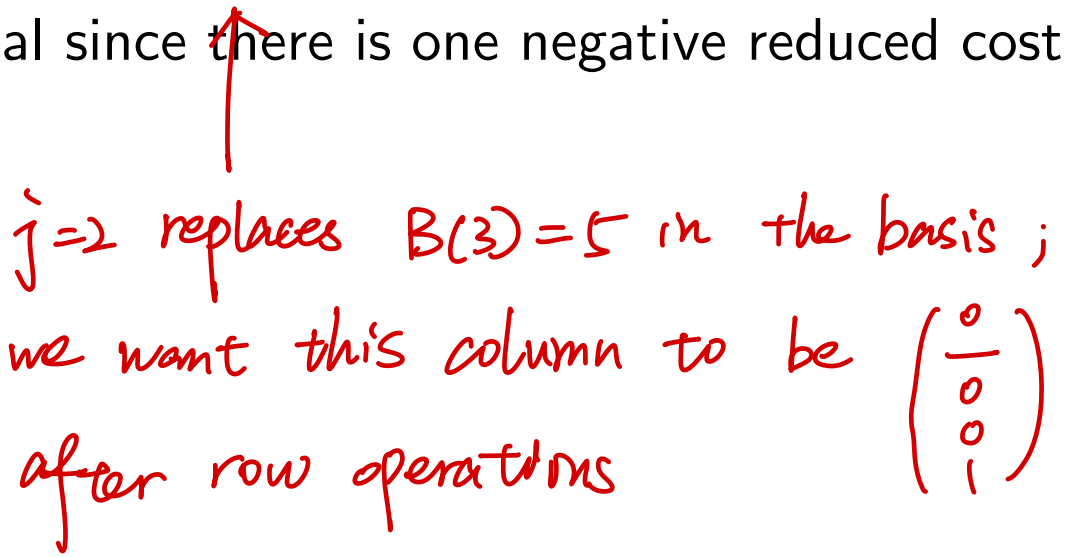
The tableau becomes:

B	0	-2	1	0	0	100
1	1	0	1	0	0	100
4	0	2	0	1	0	200
B(3) 5	0	1	-1	0	1	50

200 / 2 = 100

50 / 1 = 50

- It is not optimal since there is one negative reduced cost.



Example: Iteration II

The tableau becomes:

B	0	-2	1	0	0	100
1	1	0	1	0	0	100
4	0	2	0	1	0	200
5	0	1	-1	0	1	50

- It is not optimal since there is one negative reduced cost.
- The only choice for the pivot column is column 2.
- Use MRT, the pivot row should be row 3 (outgoing basis is column 5).

Example: Iteration II

The tableau becomes:

B	0	-2	1	0	0	100
1	1	0	1	0	0	100
4	0	2	0	1	0	200
5	0	1	-1	0	1	50

- It is not optimal since there is one negative reduced cost.
- The only choice for the pivot column is column 2.
- Use MRT, the pivot row should be row 3 (outgoing basis is column 5).

We apply the same procedure:

- Add $2\times$ row 3 to the very top row, and $-2\times$ row 3 to the second row in the constraint.

Example: Iteration III

The tableau becomes:

B	0	0	-1	0	2	200
1	1	0	1	0	0	100
4	0	0	2	1	-2	100
2	0	1	-1	0	1	50

x_4

B(2)

100/1 = 100

100/2 = 50

- It is still not optimal since there is one negative reduced cost.

do row operations to make it $\begin{pmatrix} 0 \\ 0 \\ 1 \\ 0 \end{pmatrix}$

Example: Iteration III

The tableau becomes:

B	0	0	-1	0	2	200
1	1	0	1	0	0	100
4	0	0	2	1	-2	100
2	0	1	-1	0	1	50

- It is still not optimal since there is one negative reduced cost.
- The only choice for the pivot column is column 3.
- Use MRT, the pivot row should be row 2 (outgoing basis is column 4).

Example: Iteration III

The tableau becomes:

B	0	0	-1	0	2	200
1	1	0	1	0	0	100
4	0	0	2	1	-2	100
2	0	1	-1	0	1	50

Current BFS: $\begin{pmatrix} 100 \\ 50 \\ 0 \\ 100 \\ 0 \end{pmatrix}$

- It is still not optimal since there is one negative reduced cost.
- The only choice for the pivot column is column 3.
- Use MRT, the pivot row should be row 2 (outgoing basis is column 4).

We apply the same procedure again:

- Divide row 2 by 2, then add $1\times$ row 2 to the very top row, add $-1\times$ row 2 to the first row in the constraint, add $1\times$ row 2 to the last row.

Example: Iteration IV

The tableau becomes:

B	0	0	0	1/2	1	250
1	1	0	0	-1/2	1	50
3	0	0	1	1/2	-1	50
2	0	1	0	1/2	0	100

$x_1 = 50$
$x_3 = 50$
$x_2 = 100$
$\Big\}\ x_B$

All the reduced costs are positive now:

- Thus it is optimal!
- The optimal solution is $(50, 100, 50, 0, 0)$ with optimal value -250.

Another Example

Consider the linear optimization problem:

$$\begin{aligned} \text{minimize} \quad & -10x_1 - 12x_2 - 12x_3 \\ \text{s.t.} \quad & x_1 + 2x_2 + 2x_3 \leq 20 \\ & 2x_1 + x_2 + 2x_3 \leq 20 \\ & 2x_1 + 2x_2 + x_3 \leq 20 \\ & x_1, x_2, x_3 \geq 0 \end{aligned}$$

First, we write down the standard form:

$$\begin{array}{lccccccc} \text{minimize} & -10x_1 & -12x_2 & -12x_3 & & & & \\ \text{s.t.} & x_1 & +2x_2 & +2x_3 & +s_1 & & & = 20 \\ & 2x_1 & +x_2 & +2x_3 & & +s_2 & & = 20 \\ & 2x_1 & +2x_2 & +x_3 & & & +s_3 & = 20 \\ & x_1, & x_2, & x_3, & s_1, & s_2, & s_3 & \geq 0 \end{array}$$

Simplex Algorithm: Iteration I

We write down the initial tableau ($B = \{4, 5, 6\}$):

B	-10	-12	-12	0	0	0	0
4	1	2	2	1	0	0	20
5	2	1	2	0	1	0	20
6	2	2	1	0	0	1	20

$20/1 = 20$

$20/2 = 10$

$20/2 = 10$

- This is also in a canonical form already.

there is a tie between x_5 and x_6.

→ with smallest index rule, we choose x_5 to leave

do row operations to make it $\begin{pmatrix} 0 \\ \hline 0 \\ 1 \\ 0 \end{pmatrix}$

Simplex Algorithm: Iteration I

We write down the initial tableau ($B = \{4, 5, 6\}$):

B	-10	-12	-12	0	0	0	0
4	1	2	2	1	0	0	20
5	2	1	2	0	1	0	20
6	2	2	1	0	0	1	20

- This is also in a canonical form already.

By the smallest index rule, we choose column 1 to enter the basis. By the minimum ratio test, we have two candidates to leave the basis: 5th column (row 2) or 6th column (row 3).

Simplex Algorithm: Iteration I

We write down the initial tableau ($B = \{4, 5, 6\}$):

B	-10	-12	-12	0	0	0	0
4	1	2	2	1	0	0	20
5	2	1	2	0	1	0	20
6	2	2	1	0	0	1	20

- This is also in a canonical form already.

By the smallest index rule, we choose column 1 to enter the basis. By the minimum ratio test, we have two candidates to leave the basis: 5th column (row 2) or 6th column (row 3).

By the smallest index rule again, we choose 5th column to exit (pivot row is row 2):

- Divide each element in row 2 by 2.
- Add $10\times$ new row 2 to the top row, $-1\times$ new row 2 to the first constraint row, and $-2\times$ new row 2 to the last row.

Simple Algorithm: Iteration II

Then the tableau becomes:

B	0	-7	-2	0	5	0	100
4	0	3/2	1	1	-1/2	0	10
1	1	1/2	1	0	1/2	0	10
6	0	1	-1	0	-1	1	0

$10/\frac{3}{2} = \frac{20}{3}$

$10/\frac{1}{2} = 20$

$0/1 = 0$

degenerate case

We do the same thing:

By row operations. make this column $\begin{pmatrix} 0 \\ 0 \\ 0 \\ 1 \end{pmatrix}$

Simple Algorithm: Iteration II

Then the tableau becomes:

B	0	-7	-2	0	5	0	100
4	0	3/2	1	1	-1/2	0	10
1	1	1/2	1	0	1/2	0	10
6	0	1	-1	0	-1	1	0

Column 2 is the pivot column. By MRT, the pivot row is row 3.

Simple Algorithm: Iteration II

Then the tableau becomes:

B	0	-7	-2	0	5	0	100
4	0	3/2	1	1	-1/2	0	10
1	1	1/2	1	0	1/2	0	10
6	0	1	-1	0	-1	1	0

Column 2 is the pivot column. By MRT, the pivot row is row 3.

- Here we encounter a degenerate case with minimal ratio 0.
- It means that in this pivoting, we can not strictly improve the objective value.
- But we can still proceed as normal (no cycle will occur if we use the Bland's rule):
  - We add $7\times$ row 3 to the top row, $-3/2\times$ row 3 to the first constraint row and $-1/2\times$ row 3 to the second constraint row.

Simplex Algorithm: Iteration III

Then the tableau becomes:

B	0	0	-9	0	-2	7	100
4	0	0	5/2	1	1	-3/2	10
1	1	0	3/2	0	1	-1/2	10
2	0	1	-1	0	-1	1	0

$10/\frac{5}{2} = 4$

$10/\frac{3}{2} = \frac{20}{3}$

Simplex Algorithm: Iteration III

Then the tableau becomes:

B	0	0	-9	0	-2	7	100
4	0	0	5/2	1	1	-3/2	10
1	1	0	3/2	0	1	-1/2	10
2	0	1	-1	0	-1	1	0

We choose column 3 to enter the basis. By MRT, the pivot row is row 1 (column 4 leaving basis).

Simplex Algorithm: Iteration III

Then the tableau becomes:

B	0	0	-9	0	-2	7	100
4	0	0	5/2	1	1	-3/2	10
1	1	0	3/2	0	1	-1/2	10
2	0	1	-1	0	-1	1	0

We choose column 3 to enter the basis. By MRT, the pivot row is row 1 (column 4 leaving basis).

- We multiply $2/5$ to each number in row 1, then add $9\times$ row 1 to the top row, $-3/2\times$ row 1 to the second constraint row and $1\times$ row 1 to the last row.

Simplex Algorithm: Iteration IV

Then the tableau becomes:

B	0	0	0	18/5	8/5	8/5	136
3	0	0	1	2/5	2/5	-3/5	4
1	1	0	0	-3/5	2/5	2/5	4
2	0	1	0	2/5	-3/5	2/5	4

This is optimal since all reduced costs are non-negative. The optimal solution is $(4, 4, 4, 0, 0, 0)$ with optimal value -136.

The Two-Phase Method & Simplex Tableau

Two-Phase Method in Simplex Tableau

For the simplex tableau, when there is no obvious initial basic feasible solution, we can use the two-phase method.

Phase I:

For auxiliary problem, we start from BFS $\begin{pmatrix} x \\ y \end{pmatrix} = \begin{pmatrix} 0 \\ b \end{pmatrix}$

- The auxiliary problem is not in a canonical form: Although the constraint contains an identity matrix (which is the initial basis), the corresponding objective coefficients are not 0's!
- We need to compute the top row (reduced costs):
  - For basic part, the reduced costs are 0.
  - For nonbasic part, $\bar{c}_j = c_j - \boldsymbol{c}_B^\top \boldsymbol{A}_B^{-1} \boldsymbol{A}_j = -\mathbb{1}^\top \boldsymbol{A}_j$, so the jth reduced cost is the negative of the sum of that column.
- This also applies to the initial objective value, which equals the negative of the sum of the right hand side vector.

Two-Phase Method in Simplex Tableau

Suppose we have finished the Phase I simplex tableau operations, resulting in an optimal solution $(\boldsymbol{x}^*, 0)$ with optimal value 0.

To carry out Phase II in the simplex tableau:

1. Drop all the columns associated with the auxiliary variables.
2. Change the top row to the reduced costs corresponding to the original problem by using the reduced cost formula:

$$\bar{c}_j = c_j - \boldsymbol{c}_B^\top \boldsymbol{A}_B^{-1} \boldsymbol{A}_j$$

   and compute the current (negative) objective value.
3. Continue the simplex tableau to solve the original problem.

If the optimal solution of the auxiliary problem is degenerate, then replace the basic indices of included auxiliary variables with indices of original variables to form a basis.

Example

$$
\begin{array}{lrrrrcl}
\text{minimize} & x_1 & +x_2 & +x_3 & & & \\
\text{subject to} & x_1 & +2x_2 & +3x_3 & & = & 3 \\
 & & -4x_2 & -9x_3 & & = & -5 \\
 & & & +3x_3 & +x_4 & = & 1 \\
 & x_1, & x_2, & x_3, & x_4 & \geq & 0
\end{array}
$$

First, make $\boldsymbol{b}$ positive and construct the auxiliary problem:

$y = \begin{pmatrix} x_5 \\ x_6 \\ x_7 \end{pmatrix}$

$$
\begin{array}{lrrrrrrrcl}
\text{minimize} & & & & & x_5 & +x_6 & +x_7 & & \\
\text{subject to} & x_1 & +2x_2 & +3x_3 & & +x_5 & & & = & 3 \\
 & & 4x_2 & +9x_3 & & & +x_6 & & = & 5 \\
 & & & +3x_3 & +x_4 & & & +x_7 & = & 1 \\
 & x_1, & x_2, & x_3, & x_4, & x_5, & x_6, & x_7 & \geq & 0
\end{array}
$$

start from $\begin{pmatrix} 0 \\ 0 \\ 0 \\ 3 \\ 5 \\ 1 \end{pmatrix}$

Example Continued

Construct the initial tableau for the auxiliary problem

B	-1	-6	-15	-1	0	0	0	-9
5	1	2	3	0	1	0	0	3
6	0	4	9	0	0	1	0	5
7	0	0	3	1	0	0	1	1

Carry out the simplex method. Iteration 1:

B	0	-4	-12	-1	1	0	0	-6
1	1	2	3	0	1	0	0	3
6	0	4	9	0	0	1	0	5
7	0	0	3	1	0	0	1	1

Example Continued

Iteration 2:

B	0	0	-3	-1	1	1	0	-1
1	1	0	-3/2	0	1	-1/2	0	1/2
2	0	1	9/4	0	0	1/4	0	5/4
7	0	0	3	1	0	0	1	1

Iteration 3:

B	0	0	0	0	1	1	1	0
1	1	0	0	1/2	1	-1/2	1/2	1
2	0	1	0	-3/4	0	1/4	-3/4	1/2
3	0	0	1	1/3	0	0	1/3	1/3

This is optimal for the auxiliary problem. $\boldsymbol{x} = (1, 1/2, 1/3, 0)$ is a BFS for the original problem ($B = \{1, 2, 3\}$).

Example Continued

B	0	0	0	0	1	1	1	0
1	1	0	0	1/2	1	-1/2	1/2	1
2	0	1	0	-3/4	0	1/4	-3/4	1/2
3	0	0	1	1/3	0	0	1/3	1/3

$x^* = \begin{pmatrix} 1 \\ \frac{1}{2} \\ \frac{1}{3} \\ 0 \end{pmatrix}$

We drop all the columns for auxiliary variables. We then recompute the reduced cost for the original problem for $B = \{1, 2, 3\}$:

$$\bar{\boldsymbol{c}}^\top = \boldsymbol{c}^\top - \boldsymbol{c}_B^\top \boldsymbol{A}_B^{-1} \boldsymbol{A} = (0, 0, 0, -1/12).$$

We also need to compute the current objective value: $11/6$.

Now the simplex tableau becomes:

B	0	0	0	-1/12	-11/6
1	1	0	0	1/2	1
2	0	1	0	-3/4	1/2
3	0	0	1	1/3	1/3

Example Continued

Then we continue from the new simplex tableau:

B	0	0	0	-1/12	-11/6
1	1	0	0	1/2	1
2	0	1	0	-3/4	1/2
3	0	0	1	1/3	1/3

The next pivot:

B	0	0	1/4	0	-7/4
1	1	0	-3/2	0	1/2
2	0	1	9/4	0	5/4
4	0	0	3	1	1

This is optimal. The optimal solution is $\boldsymbol{x} = (1/2, 5/4, 0, 1)$. The optimal value is $7/4$.

Simplex Method Summary

We have completed our discussions of the simplex method.

- The idea of simplex method (search among the BFS and neighbors) and its justifications.
- The algebraic procedures.
- The simplex tableau.
- Other issues about the simplex method: initialization, degeneracy, etc.

Today's lecture

- The Simplex Tableau
- Two-phase Method with Simplex Tableau

Next lecture

- Complexity
- Duality

Questions?